

David's Theory of Perceptual Abstraction Applied to Economics

A relational account of economic change, information, and contextual choice

Anonymous Working Paper · 8 October 2026

Abstract. This paper develops a compact economic formalization of David's Theory of Perceptual Abstraction. The central claim is that economic instability is neither solely a property of an agent nor solely a property of an event: it is generated by a relation whose capacity to produce change interacts with an agent's sensitivity to change. We define relational economic capacity, agent sensitivity, and perceptual abstraction as measurable objects. Shannon information theory supplies an operational language for uncertainty reduction and signal value. A quantum-inspired analogy is used carefully: contextual probability and order effects are imported as mathematical metaphors for incompatible frames, without claiming microscopic quantum causation. The paper states and proves propositions on amplification, threshold transitions, and information-sensitive choice; gives a worked market example; and proposes a falsifiable empirical design using panel data and randomized information treatments. The resulting framework is simple enough to estimate, yet rich enough to generate distinctive predictions.

Keywords: perceptual abstraction; relational instability; economic sensitivity; Shannon information; contextual choice; bounded rationality; causal inference

1. Motivation and conceptual translation

Economic theory often separates the object that changes from the subject that responds. Prices, technologies, institutions, and shocks are modeled as causes; firms, households, and investors are modeled as responders. David's theory suggests a different starting point: what matters is the relation through which a difference becomes perceptible and consequential. An identical price movement may be negligible for one agent, decisive for another, and transformative for a market when many agents are coupled.

We therefore distinguish two channels. *Relational capacity* measures how strongly a situation can generate an economically relevant change. *Agent sensitivity* measures how strongly a given agent converts a perceived difference into action. Their product is an interaction intensity, while their composition through feedback determines instability. This preserves economic direction: capacity belongs to the opportunity or event as situated in a network, sensitivity belongs to the decision-maker, and the observable effect belongs to the relation.

2. Formal primitives

Let time be discrete, $t=0,1,\dots$, and let an economic relation be $R=(i,j,c)$, connecting agent i to object, counterparty, or context j under condition c . Let x_t be a state vector (prices, quantities, beliefs, liquidity, or institutional variables), and let $y_t=g(R,x_t)$ be the perceptual signal available to i . A baseline reference y_0 defines the difference $d_t=y_t-y_0$.

Define relational economic capacity $C_R(t)$ as the norm of the state transition attributable to the relation, normalized by available resources:

$$C_R(t) = || E[x_{t+1} - x_t \mid \text{do}(R, c)] || / (1 + ||x_t||), \quad C_R \geq 0.$$

Define agent sensitivity $S_i(t)$ as the local slope of action with respect to perceived difference, allowing saturation:

$$S_i(t) = || \partial a_i(t) / \partial d_t || \cdot q_i(t), \quad 0 \leq q_i \leq 1.$$

Here a_i is an action vector (demand, investment, search, exit), and q_i captures attention, liquidity, or execution constraints. The relational instability index is

$$h_D(R, i, t) = C_R(t) S_i(t) / (1 + \kappa ||d_t||), \quad \kappa \geq 0.$$

The denominator makes the index compatible with diminishing marginal response. High h_D means that a small, economically available difference is both capable of propagating and likely to be acted upon.

3. Shannon information and abstraction

Let Z be a finite-valued economic state and Y the signal observed by an agent. Shannon entropy is $H(Z) = -\sum_z p(z) \log p(z)$. The information supplied by Y about Z is $I(Z; Y) = H(Z) - H(Z|Y)$. We define perceptual abstraction as a map $A_\theta: Y \rightarrow U$ that retains decision-relevant distinctions while discarding detail. The parameter θ controls granularity.

A useful economic measure is the information-adjusted sensitivity

$$S_i^I(t) = S_i(t) \cdot I(Z; Y_i) / (H(Z) + \epsilon), \quad \epsilon > 0.$$

This equation says that sensitivity is not merely psychological. It is economically effective when the perceptual channel carries information about the state that matters for action. A highly reactive agent receiving an uninformative signal can produce noise; a moderately reactive agent receiving a highly informative signal can produce efficient adjustment. Abstraction is therefore a compression problem with a welfare trade-off: excessive compression hides causal distinctions, while insufficient compression overloads attention.

4. Dynamic system and contextual decision analogy

A minimal coupled system is

$$x_{t+1} = x_t + \alpha C_R(t) a_t + u_t, \quad a_t = \sigma[\beta S_i(t) d_t + \gamma m_t],$$

where α is transmission, β is response gain, γm_t is a contextual term, σ is a bounded choice function, and u_t is an exogenous disturbance. Linearizing around a fixed point gives $\Delta x_{t+1} \approx J \Delta x_t$. Local instability occurs when the spectral radius $\rho(J) > 1$.

The quantum-inspired analogy concerns contextual probability, not physical quanta. Let a decision be represented by a unit vector $|\psi\rangle$ in a two-dimensional state space, and let a context k define a projector Π_k . Choice probability is $p(k) = \langle \psi | \Pi_k | \psi \rangle$. Two contexts need not commute: $\Pi_1 \Pi_2 \neq \Pi_2 \Pi_1$. Hence asking about risk before asking about fairness can alter the probability of choosing a product, even when objective facts are unchanged. The analogy predicts order effects and interference-like deviations from a single classical ranking, while making no claim that markets are quantum systems.

5. Propositions

Proposition 1 (Relational amplification). Suppose C_R , S_i , and α are nonnegative and σ is differentiable with local derivative σ' satisfying $0 < \sigma' \leq 1$. In a scalar linearization, the fixed point is locally unstable whenever $\alpha\beta C_R S_i \sigma' > 1$.

Proof. The update derivative is $J=1+\alpha\beta C_R S_i \sigma'$ for a level variable, or $J=\alpha\beta C_R S_i \sigma'$ for a mean-reverting deviation variable. In the latter economically relevant representation, $|J|>1$ is exactly the stated condition. Since all factors are nonnegative, the threshold is crossed by increasing capacity, sensitivity, transmission, or response gain. ■

Proposition 2 (Information can stabilize or destabilize). Let $S^I = S \cdot I/H$. If information changes only I and the local feedback multiplier is $M=\alpha\beta C_R S^I \sigma'$, then increasing I raises instability monotonically when $C, S, \alpha, \beta, \sigma'$ are fixed. However, if abstraction also reduces noise variance $v(I)$ and σ' is decreasing in perceived variance, the total effect can be non-monotone.

Proof. Under fixed σ' , $\partial M/\partial I = \alpha\beta C_R S^I \sigma' / H > 0$. If $\sigma' = \phi(v(I))$ with $\phi' < 0$ and $v' < 0$, the product derivative contains a positive information term and a negative variance term. Their signs may offset, producing an interior maximum. ■

Proposition 3 (Context order effect). Let Π_1, Π_2 be noncommuting projectors and $|\psi\rangle$ a choice state. Then sequential probabilities $p(1 \rightarrow 2) = \|\Pi_2 \Pi_1 |\psi\rangle\|^2$ and $p(2 \rightarrow 1) = \|\Pi_1 \Pi_2 |\psi\rangle\|^2$ can differ.

Proof. Equality would require equality of the two quadratic forms for the given state. Noncommutation permits distinct products and therefore distinct norms for an open set of $|\psi\rangle$. ■

6. Worked example: a liquidity-sensitive market

Consider a small-business credit market. A lender receives a repayment signal d_t from a borrower network. Network centrality and collateral determine C_R ; cash buffer and attention determine S_i . Let $C=0.8$, $S=1.5$, $\alpha\beta\sigma'=0.95$, and normalized information $I/H=0.7$. Then the effective multiplier is $M=0.95 \times 0.8 \times 1.5 \times 0.7 = 0.798$, below one: shocks decay. A public dashboard raises I/H to 0.95 while also making lenders more reactive; $M=1.083$, above one: a small delinquency signal can trigger synchronized credit withdrawal. The theory predicts that the same dashboard can improve forecast accuracy and worsen systemic stability, depending on whether it increases useful information faster than it reduces interpretive variance.

7. Falsifiable empirical design

The framework can be tested with a panel of firms, lenders, or households observed before and after exogenous information releases. Estimate C_R from a first-stage response of local outcomes to a relation-specific shock, and estimate S_i from heterogeneous action elasticities to the same perceived difference. Use randomized message framing to vary abstraction: raw statistics, an aggregate index, or a contextual narrative. The primary outcome is the interaction coefficient in

$$\Delta a_{it} = \delta d_{it} + \eta C_{Rt} + \theta S_i + \lambda (C_{Rt} \times S_i) + X_{it}' \gamma + \mu_i + \tau_t + \varepsilon_{it}.$$

The central prediction is $\lambda > 0$ for amplification, with a threshold in aggregate volatility when estimated M crosses one. A contextual-choice module randomizes question order; nonzero order effects after controlling for objective information support the contextual analogy. Falsification tests include placebo orderings, pre-treatment balance, and outcomes outside the information channel. Heterogeneous effects by liquidity and attention provide a direct test of sensitivity rather than a generic shock response.

8. Limitations and scope

The index h_D is a disciplined abstraction, not a universal law. Capacity is difficult to identify when networks and expectations co-evolve. Sensitivity may be endogenous to prior shocks, and Shannon information depends on a chosen state space. The quantum-inspired representation is useful for contextual probability and order effects, but it should not be interpreted as evidence of quantum physics in cognition or markets. Finally, a scalar threshold can hide multidimensional feedback, strategic anticipation, and institutional constraints. These limits define empirical work rather than invalidate the framework.

9. Conclusion

David's Theory of Perceptual Abstraction becomes economically precise when change is modeled as relational. Capacity describes how a relation can move an economic state; sensitivity describes how an agent converts a perceived difference into action; information and abstraction determine which differences enter the decision channel. Their interaction generates a measurable instability index, dynamic thresholds, and distinctive predictions about feedback and context. The theory is valuable precisely because it can fail: data can reject amplification, information can reduce rather than increase volatility, and order effects can disappear. That makes the proposal a research program rather than a metaphor.

References

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